Timestamp Analysis (2)

1 Functional Requirement: Timestamp Analysis 0

1 Functional Requirement: Timestamp Analysis

Unique Id: LD_Req-76464 v2

ASIL: Safety-related no ASIL

The system shall convert a 64-bit nanosecond timestamp to milliseconds and extract the least significant 12 bits using a bitmask.

Camera Id and TimeStamp Insertion (2)

1 Functional Requirement: Camera Id and TimeStamp Insertion 0

1.1 Sub Statement: Timestamp Description 0

1 Functional Requirement: Camera Id and TimeStamp Insertion

Unique Id: LD_Req-76465 v2

ASIL: Safety-related no ASIL

Colour coded timestamp and Camera ID signature shall be added to each image frame by the HV Image Stream at the right corner.

Each colour represent 4 bits, or a single hexadecimal digit. These colours were selected in a way that are both visually and value-wise distinctive.

The diagram below present how each hex value is represented as colours in RGB format.

New colour coding		
0	6D2400	
1	009100	
2	DA9100	
3	6DFF00	
4	FF0024	
5	00246D	
6	6D916D	
7	00FF6D	
8	DAFF6D	
9	910091	
10	FF6D91	
11	0091DA	
12	6DFFDA	
13	2400FF	
14	FF00FF	
15	916DFF	
16	FFDAFF	replace black

The signature is a 16-pixel wide column replacing the image's rightmost part. From top to bottom, the signature is represented by 8 blocks of 16*16 pixel as follows:

- 4 Bits (1 Block) to represent the camera ID
- 12 Bits (3 blocks) to represent the timestamp using the extrated time stamp from Timestamp Analysis. Encoding data of is in Little Endian format (MSB to LSB- top to bottom)
- Padding – 4 blocks

Note:

The HV Image stream shall add the camera ID and timestamp information per camera view in the order of which the camera view is added.



1.1 Sub Statement: Timestamp Description

Unique Id: LD_Req-76466 v1

ASIL: To Be Selected



Camera Meta Data (1)

3 Functional Requirement: Camera Meta Data

Unique Id: LD_Req-76467 v1

ASIL: To Be Selected

The system shall provide camera meta data, including the following categories:

- System-Level Metadata: Mosaic algorithm version or configuration/view ID
- Timing and Sync Metadata: Frame sync status (e.g., dropped frames, latency), clock drift, sync error
- Diagnostic Metadata: Last successful mosaic rendering time"